IronKaggle: King County Houses

A predictive machine learning project by **Feliciu & Enrique** — leveraging **21,613 records** and **21 distinct features** to deliver accurate housing price estimations through advanced EDA, geospatial modeling, and ensemble methods.

21,613 Property Records	21 Key Variables
27 Engineered Features	

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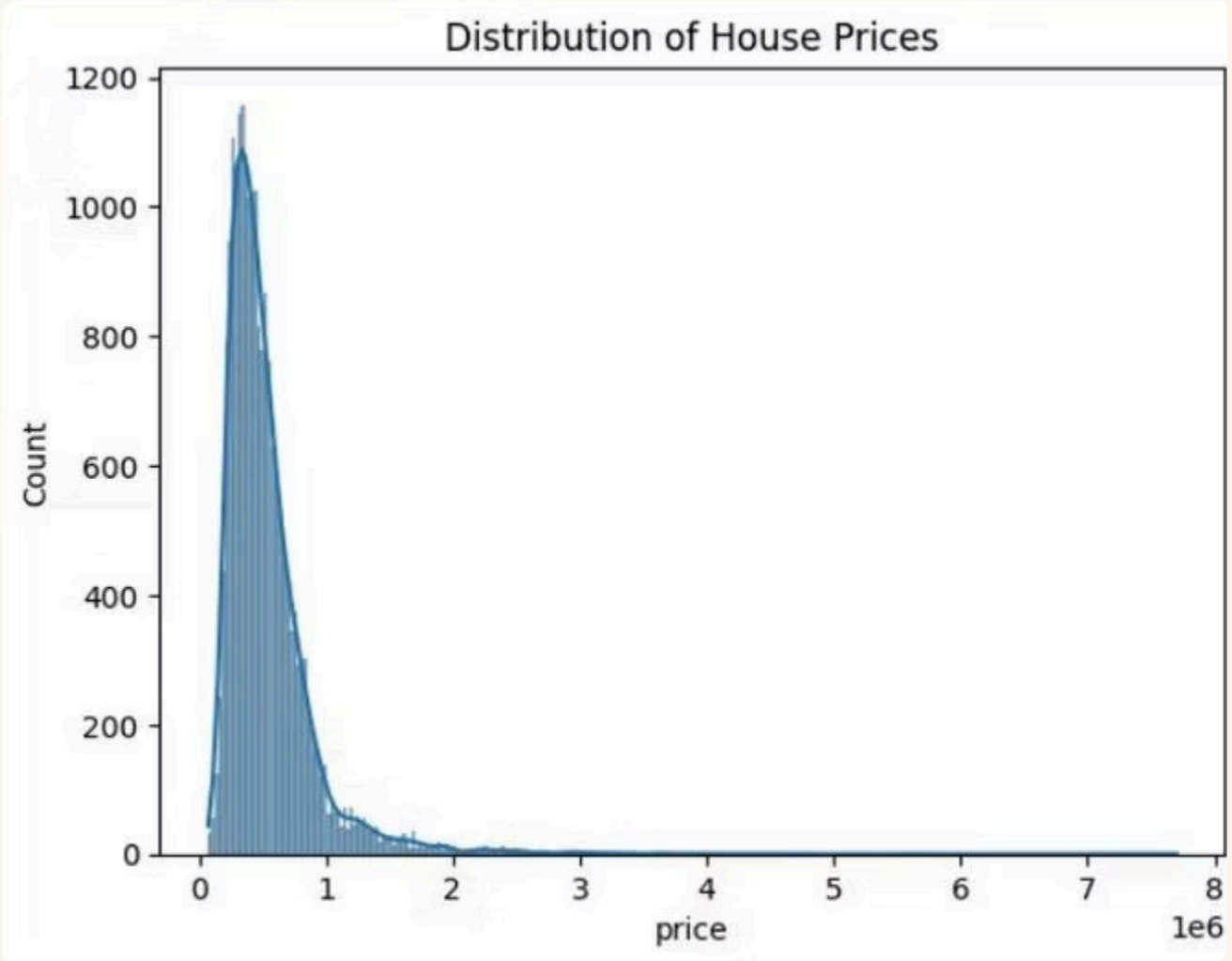
EDA

Dataset Overview & Price Distribution



Key Statistics

21,613 Entries

Zero null values — clean dataset

Right-Skewed

Log transformation applied for model stability

Outliers Removed

e.g., a property listed with 33 bedrooms

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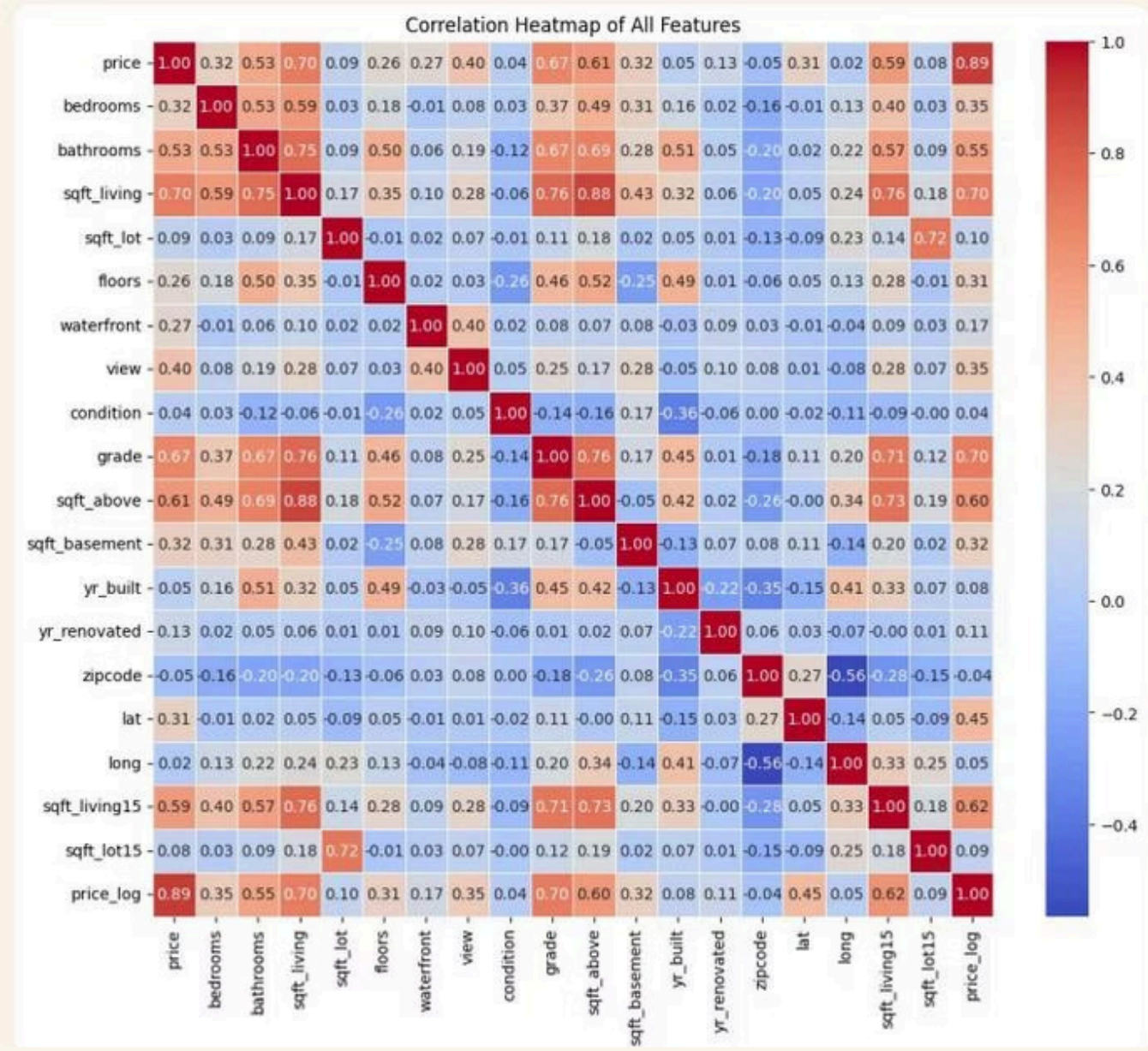
CORRELATION

Top Predictors of Price



sqft_living — 0.70

Strongest physical predictor of price

grade — 0.67

Quality construction correlates strongly with price

bathrooms — 0.53

Reliable proxy for size and luxury

view — 0.40

Good secondary value indicator

Engineering insight: yr_built (0.05) and yr_renovated (0.13) are weak raw — combine into house_age and is_renovated.

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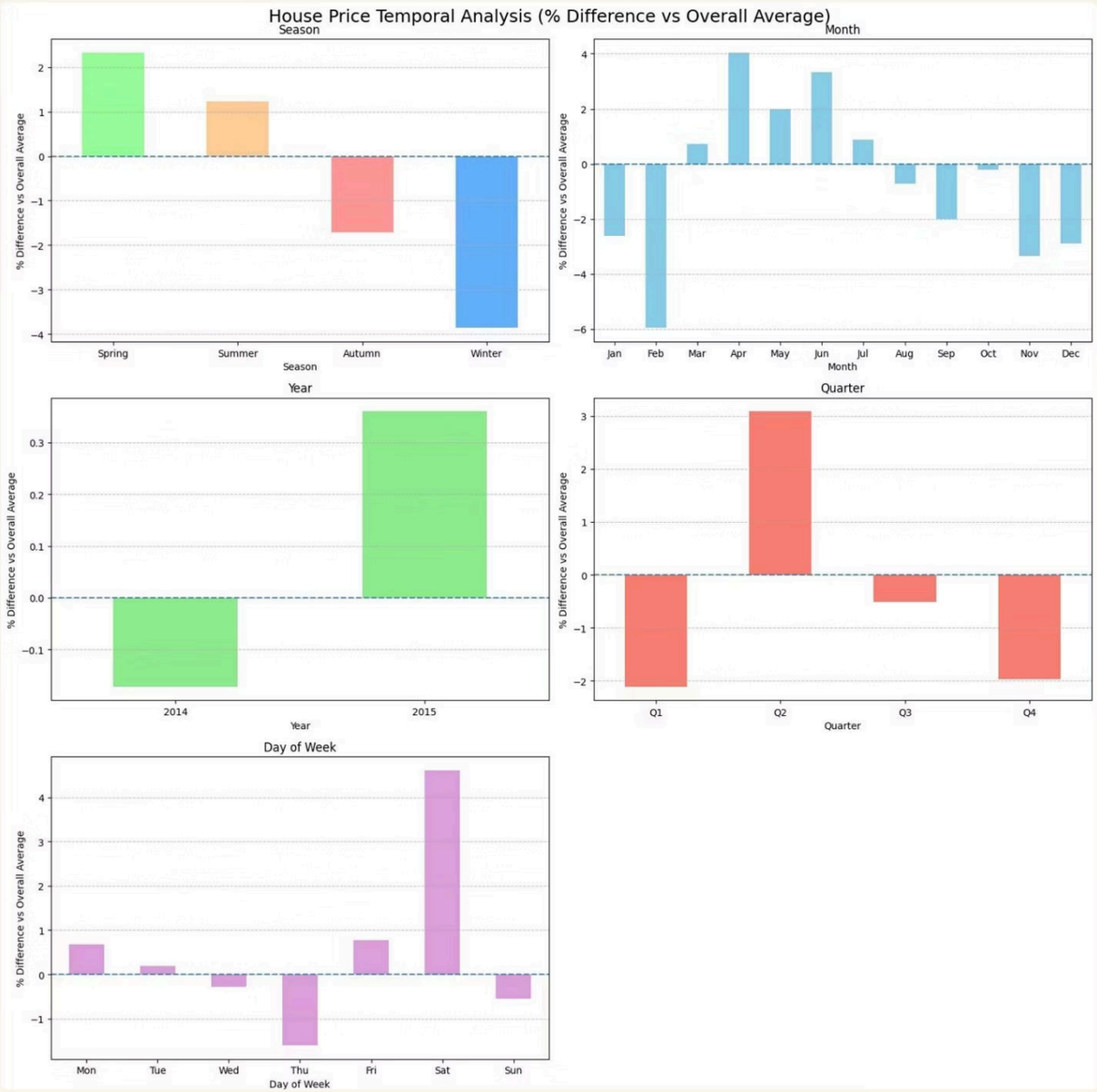
TEMPORAL ANALYSIS

Temporal Drivers of Valuation



🌸 Spring Leads

Prices peak ~6% above average; Winter dips ~2%.

📅 Q2 Dominance

Q2 consistently outperforms Q1 and Q4 (+5% avg).

📅 Thursday Peak

Listings on Thursdays command ~3% premium.

📊 Conclusion

Temporal effects are statistically significant but small (<10% variance). Structural features remain primary drivers.

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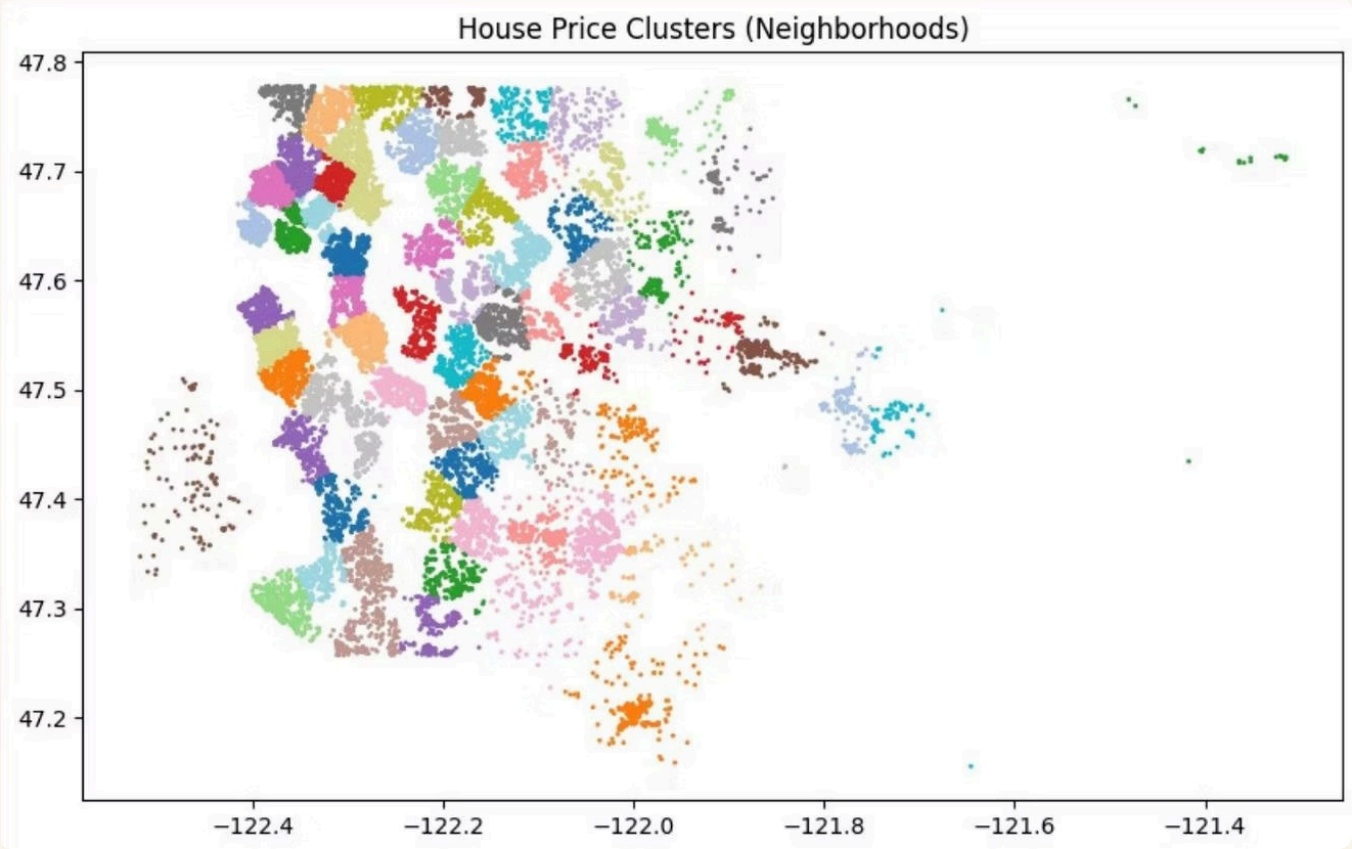
GEOSPATIAL MODELING

Geospatial Clustering Analysis



Clustering Strategy

KMeans engineered **70 price-based neighborhoods** from lat/long, mirroring the 70 zip codes. Achieved **0.72 cluster-zipcode purity**.

Model	R^2	RMSE
Zipcode	0.391	$299,987
Geo Cluster	0.359	$307,663
Lat/Long RF	**0.583**	—

Raw lat/long + Random Forest significantly outperforms discrete spatial encodings (R^2=0.583 vs 0.391). Geospatial drivers are highly non-linear.

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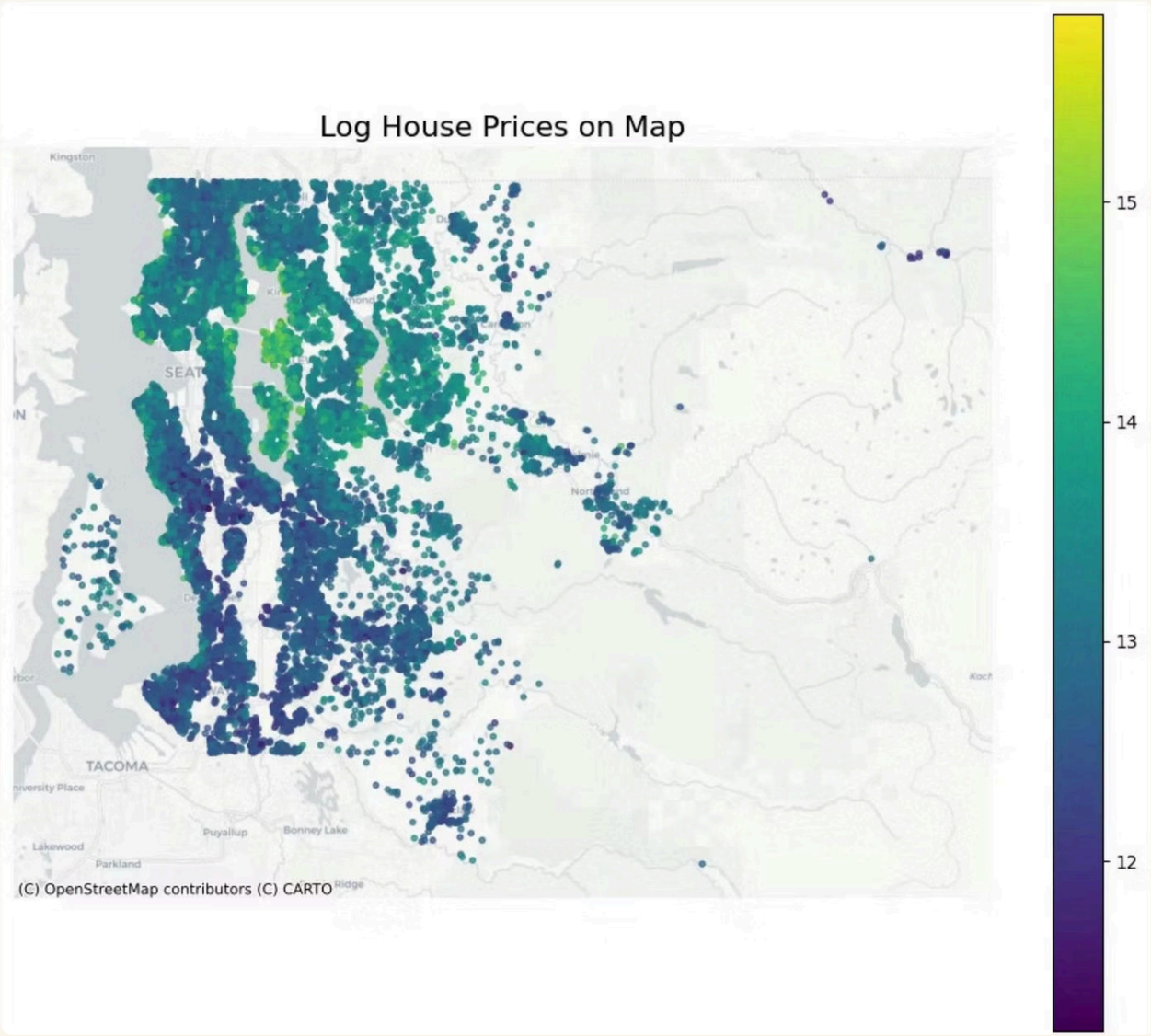
SPATIAL GRADIENT

Log House Prices Across King County



Epicenter

Seattle center (47.62°N, -122.33°W) is the primary price driver.

Distance Decay

Values decline consistently from urban core to suburbs.

70 Neighborhoods

Hyper-local segmentation captures variations that zip codes obscure.

Non-Linear

Advanced models like Random Forest are required to leverage spatial insights.

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FEATURE ENGINEERING

Feature Engineering & Time-Based Split

1

Engineer Features

house_age, is_renovated, dist_to_center, neighborhood_group

2

Target-Encode Zipcode

Avoids data leakage from high-cardinality categorical

3

Extract Temporal

season, quarter, month, day from date field

4

Time-Based Split

Preserves temporal order — no future data leaks into training

Final feature set: **27 features** spanning size, location, structure, quality, age, and temporal variables.

Index(['bedrooms', 'bathrooms', 'sqft_living', 'sqft_lot', 'floors', 'waterfront', 'view', 'condition', 'grade', 'sqft_above', 'sqft_basement', 'lat', 'long', 'sqft_living15', 'sqft_lot15', 'is_renovated', 'house_age', 'house_age_renovated', 'year', 'month', 'day', 'dayofweek', 'quarter', 'lat_long', 'lat_sq', 'long_sq', 'zipcode_te'], dtype='str') 27

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MODEL SELECTION

Cross-Validation Results & LightGBM Tuning

CV Results (TimeSeriesSplit, n=5)

Model	MAPE	R^2	RMSE
LightGBM ⭐	**11.79%**	**0.892**	**$120K**
Gradient Boosting	12.32%	0.886	$123K
Random Forest	12.52%	0.829	$151K
Linear Regression	14.66%	0.833	$149K
Decision Tree	16.58%	0.740	$186K

Best LightGBM Params

n_estimators: **2000** · learning_rate: **0.02**

num_leaves: **20** · subsample: **1.0**

colsample_bytree: **0.8** · reg_alpha/lambda: **0.1**

Final Test Performance

R^2 **0.894** · MAPE **12.2%** · RMSE **$119K**

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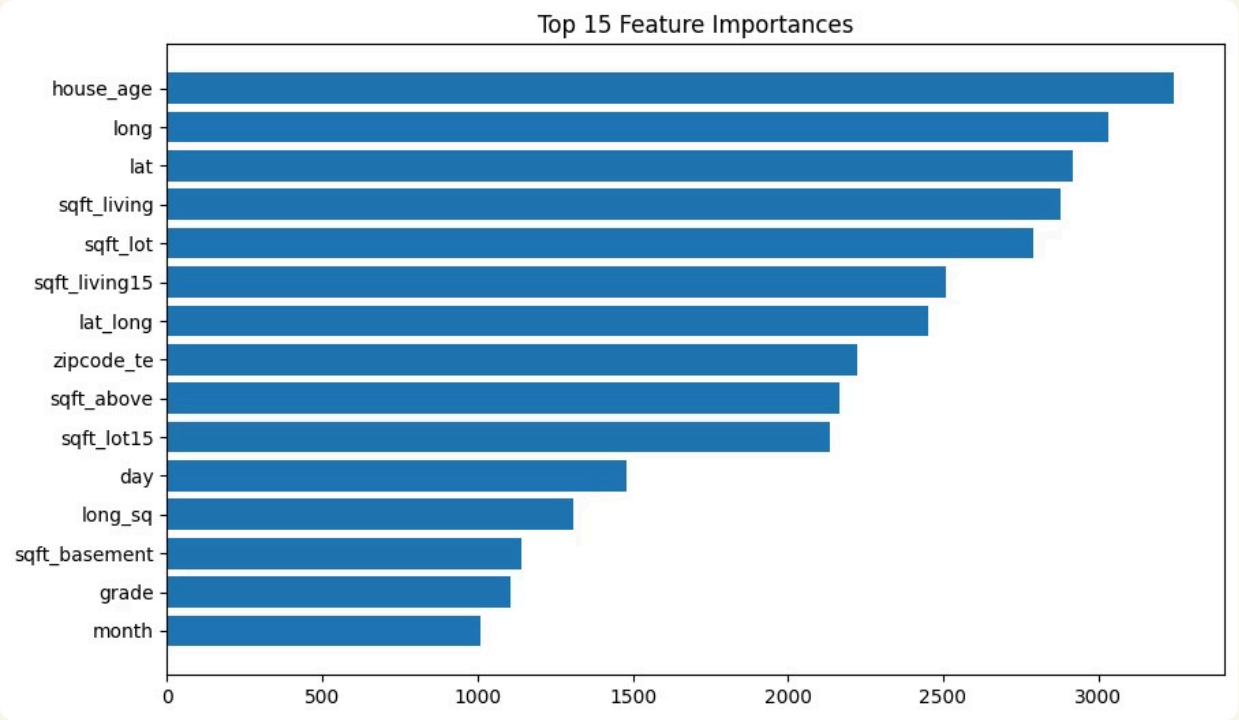
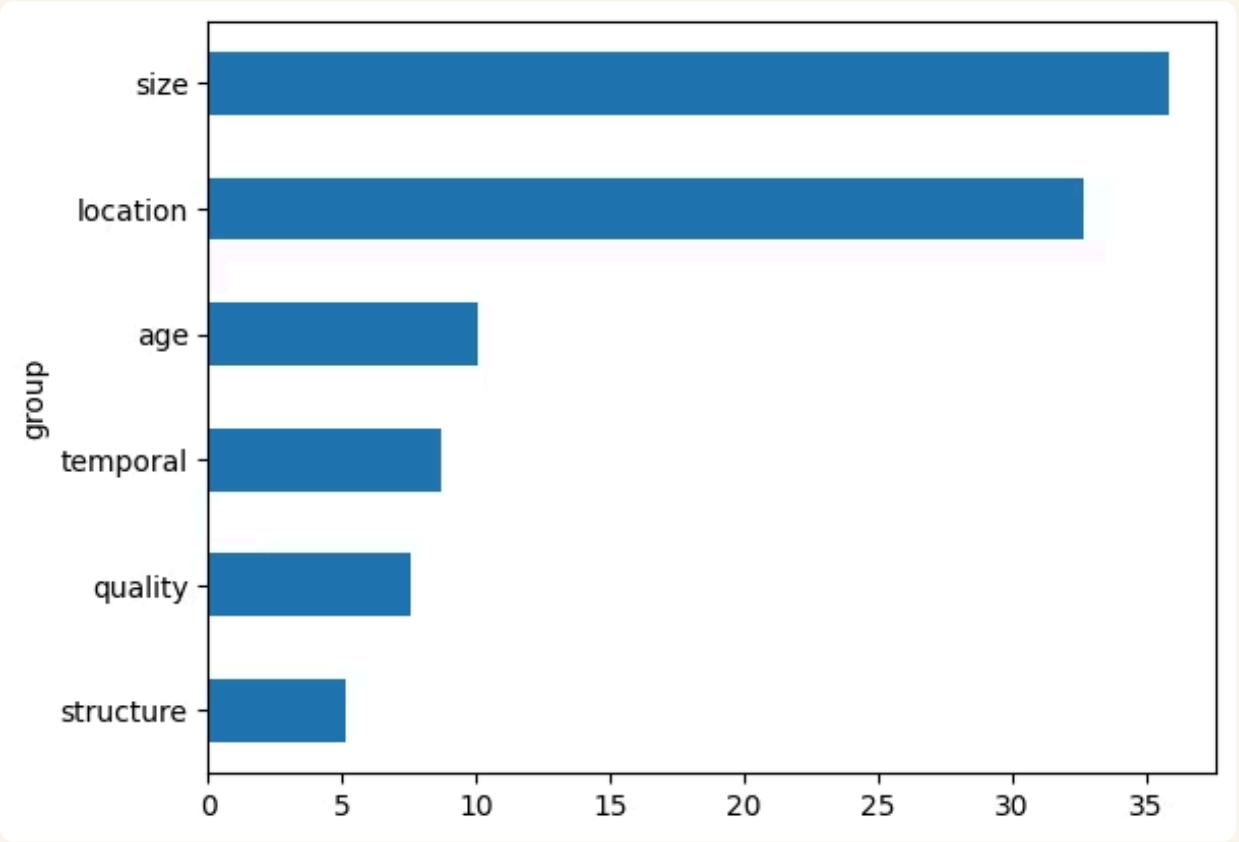
FEATURE IMPORTANCE

What Drives the Model?





📐 Size — 36%

sqft_living, sqft_lot, sqft_above, sqft_basement

📍 Location — 33%

lat, long, lat_long, zipcode_te

🏚 Age — 10%

house_age is the #1 individual feature (score: 3,245)

📅 Temporal — 9%

year, month, day, quarter

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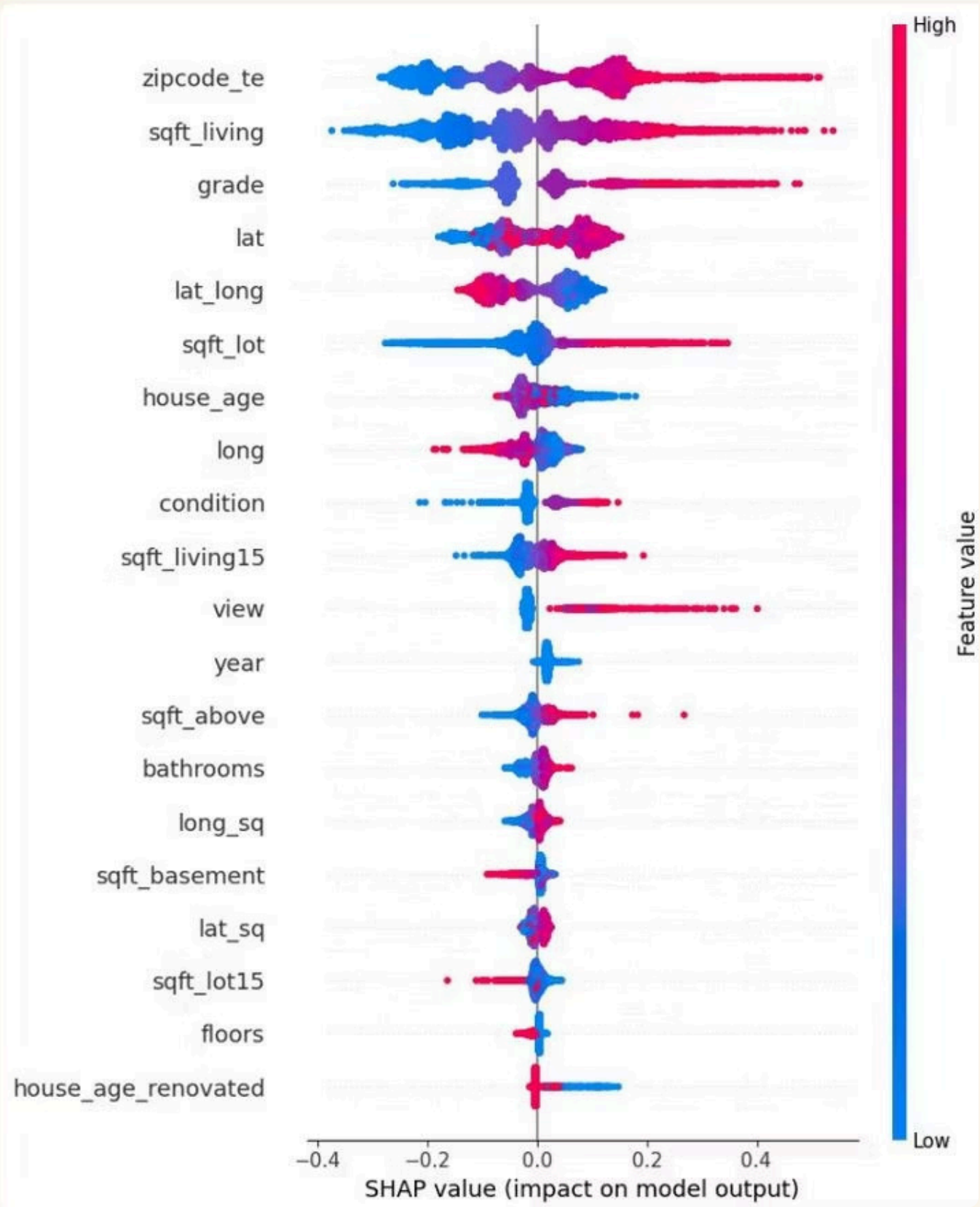
SHAP EXPLAINABILITY

SHAP Insights & Final Ensemble



Key SHAP Drivers

zipcode_te & sqft_living

Top SHAP drivers — high values strongly push price predictions up

grade & lat

High grade and favorable latitude both increase predicted price

house_age

Newer homes push predictions higher; older homes pull them down

Final Ensemble — R^2 = 0.887

LightGBM	60%
GradientBoosting	25%
RandomForest	15%

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